

13

Precision for Planning

Precision is indicated by the width of a CI. Actually it's MOE, the half-width of the CI, or one arm of the CI that's our measure of precision. Higher precision is signaled by a shorter CI, and short CIs are good news. Precision is a highly valuable idea, but not sufficiently recognized or used. It usually contributes to what in Chapter 12 I called the *informativeness* of an experiment.

Most of this book so far has been about using CIs to report and interpret results, and one valuable approach to interpreting a CI focuses on MOE as a measure of precision. This chapter considers a further use of precision—in the planning of experiments. To carry out a precision analysis for planning, we first select a target CI width, and then the analysis tells us what sample size is likely to give CIs no wider than that target. In general, researchers have not yet made precision a central part of their research planning. When they do, they'll no longer need to use statistical power. Precision for planning has the great advantage that it uses, before the experiment, CI concepts and judgments that correspond closely with those we use after the experiment for interpretation. This contrasts strongly with power, which really only applies before the experiment.

Here's the agenda for this chapter:

- Precision as arm length
- Precision for research planning, for three experimental designs
- Precision for planning using ESCI
- Precision with assurance: finding N so we can be reasonably assured our CI won't be wider than the target width we've chosen
- Precision for planning using Cohen's d
- Ways to increase precision

Precision as Arm Length

Consider any of the HEAT (Hot Earth Awareness Test) experiments we discussed in Chapter 12. To report our main result we'd no doubt use a CI,

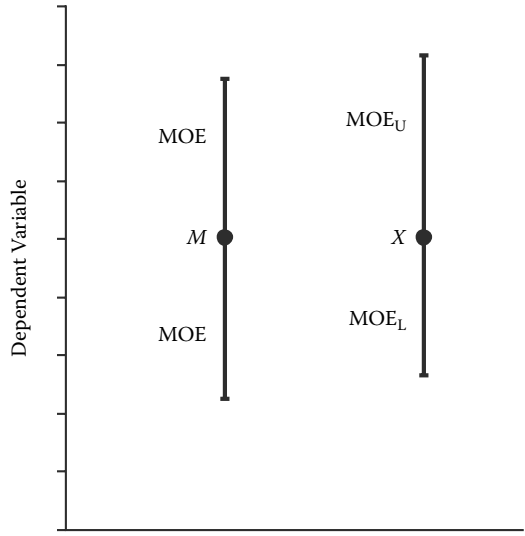


FIGURE 13.1

On the left is a symmetric CI on a sample mean M . On the right is an asymmetric CI on some ES measure X , the point estimate of the corresponding population ES. The upper and lower arms of the intervals are labeled.

which we'd display as the graphic on the left in Figure 13.1. I've labeled the arms of the CI as MOE because my focus in this chapter is on precision, as pictured by arm length and measured by MOE. Whenever we include a CI

I define *precision* as the arm length of a CI, and refer to it as MOE. in a figure we're using precision to report our results. Precision is also the basis for our third approach to interpreting a CI,

as we discussed in Chapter 3—see also Table 5.1. Precision indicates the maximum likely error of estimation, although, of course, larger errors are possible—our CI just may be red, meaning it does not capture μ , or whatever population parameter it's estimating. Vary the level of confidence, C , and MOE and thus precision change. Our measure of precision therefore relates to a particular C , usually 95.

Figure 13.1 shows on the left the CI on our sample mean M . This CI is symmetric and so the lower and upper arms can both be labeled as MOE. On the right is X , the point estimate of a population ES, with a CI that is asymmetric, meaning its two arms differ in length. We've already encountered in Chapter 11 asymmetric CIs on Cohen's d . In Chapter 14 we'll find that CIs on correlations and proportions are also generally asymmetric. Figure 13.1 distinguishes the two arms of the asymmetric CI by labeling the upper and lower arms as MOE_U and MOE_L , respectively. For symmetric intervals I'll use MOE for the length of either

For an asymmetric CI, our measure of precision is MOE_{avr} , which is the average of the two arm lengths.

arm. For asymmetric intervals we'll need to label the arms differently, as in Figure 13.1. For asymmetric intervals our measure of precision is simply MOE_{av} , the average length of the two arms, and so

$$MOE_{av} = (MOE_L + MOE_U)/2$$

It may feel strange to say *increased* precision gives *shorter* CI arms, or a *smaller* MOE, but that's what I will do. Watch out for possible ambiguities in what you read or write about precision. Wide CIs mean low precision, and small MOE means high precision. If necessary, add a few words to make what you mean totally clear.

Take care with language. A shorter CI means lower values of MOE, and greater precision. Larger MOE means worse precision.

Precision for Research Planning

You are investigating the effectiveness of a new reading program and wish to estimate the gain in reading age produced by 12 weeks on the program. How large an experiment do you need? Rather than asking about power to detect a true gain of a stated target size, if it exists, focus on estimation.

Target precision, or *target MOE*, is the precision we want our experiment to achieve.

How wide could a CI be and still give useful information? How short a CI arm length would you like to achieve? What's

the maximum error of estimation you can readily accept? You need to answer one of those questions by stating your *target MOE*, which is the precision you want your experiment to achieve, then you (usually meaning ESCI) can calculate what N you need. That's precision for research planning, the topic of this chapter.

Within NHST, statistical power contributes to planning, but precision can do a better job in that role. Precision often allows us to avoid NHST both for planning before the experiment, and also for data analysis and interpretation after the experiment. Precision for planning requires no specification of α and null and alternative hypotheses. Also, it allows us to think in terms of estimation before data collection as well as for reporting our research, and the judgments needed for planning correspond closely with those we use for interpretation of our results. Precision is better than power.

Given a target MOE, precision calculations tell us what N we need. Precision can thus replace power.

I'll discuss how we can set target MOE, then use this to calculate the N needed so we can expect our experiment to give precision at least that great. We specify a desired arm length, then calculate what minimum N is likely to give us a CI with arms no longer than that. I'll consider

three experimental designs: single group, two independent groups, and paired—each for the case where σ is known and we use z as the test statistic, and the more realistic case in which σ is not known and we use t . First I'll use means as our ES, then Cohen's d . There are formulas, but you can let ESCI do the work if you wish.

You may be thinking about the dance of the CIs when t is our test statistic: CI length varies from experiment to experiment. How can we say that N is adequate to give a particular target MOE if arm lengths bounce all over the place? That's an excellent question. Notice that I carefully referred above to N sufficiently large that we can *expect* our experiment to give us sufficiently high precision. The first big step, when we're using t , is to find minimum N that will *on average* give us CI arms no longer than target MOE. In other words, minimum N that's *likely* to give us CI arms no longer than target MOE. Almost 50% of experiments will give MOE greater than our target, the other experiments MOE less than our target. We can't guarantee that our single experiment will have the precision we want, but we can be sure that numerous repeated experiments will on average give at least our desired precision. Later we'll do even better when we consider *assurance*, but, for the moment, we'll consider what minimum N is likely to give arm length no greater than target MOE. Later I'll also explain why, even when we think about variation over experiments, precision is better than power. Now let's turn to the simplest cases of precision analysis.

Finding N for the Single-Group Design

Recall from Chapter 12 that the HEAT has $\mu = 50$ and $\sigma = 20$ in the reference population. Suppose you take a single sample of $N = 10$ students and find $M = 57.0$. Assuming that σ is known, you use Equation (3.2)—remember way back then?—to find the 95% CI by calculating

$$\text{MOE} = 1.96 \times \sigma / \sqrt{N} = 1.96 \times 20 / \sqrt{10} = 12.40$$

and so the CI is [44.6, 69.4]. You are a bit disappointed that MOE is so large, although you realize your sample is very small. How large a sample, you wonder, would you need to get smaller MOE, say, 10 instead of 12.4? The 10 is your choice, made by using your judgment, based on your

We express our chosen target precision as $f \times \sigma$, so f is usually a fraction, for example, $f = 0.5$.

understanding of the HEAT scale and the research context. We'll call 10 your target MOE. You fiddle with the MOE formula a bit and decide that you should express

target MOE as a fraction of σ . I'll use f for that fraction, so I can say that your target MOE is $f = 0.5$, because 10 is just half of σ . Our basic formula for target MOE is therefore

$$\text{Target MOE} = f \times \sigma \quad (13.1)$$

Think of f as a kind of z score, a number of SDs. Your fiddling with the MOE formula gave you

$$N = (1.96/f)^2 \quad (13.2)$$

For $f = 0.5$ you calculate $N = 15.4$. Round that up to 16, obtain the HEAT scores for a sample of $N = 16$ students, and use Equation (3.2) to calculate $\text{MOE} = 1.96 \times$

If σ is known, turn around the formula for MOE to calculate N that gives a particular target MOE.

$\sigma/\sqrt{N} = 1.96 \times 20/\sqrt{16} = 9.80$. Success! Your MOE is just less than your target of 10—slightly less because we rounded N up from 15.4 to 16. You are also pleased to note that we get this MOE every time: We're using z , because we're assuming that σ is known, and so MOE doesn't vary from experiment to experiment.

Therefore, if we're using a single sample and test statistic z : *To get MOE no more than half of σ we need to use N at least 16.* That's a general statement and a useful one—exactly the information we need to help us plan our study. Choose a different f and we'll almost certainly need a different N , but Equation (13.2) will give us that N . Equation (13.2) states a general relationship, which does not depend on σ : Given only f , we can calculate N .

Figure 13.2 shows 95% CIs on a sample mean, with arm lengths that are various fractions f of σ . The line segment at left marks σ , and the leftmost CI has $f = 0.5$, meaning $\text{MOE} = 0.5\sigma$, as you used previously. We found that such a CI requires $N = 16$, and this value is reported below the CI as **N using z** . The figure tells us that if we want, for example, $f = 0.2$, we need N at least 97. Yes, high precision, meaning small f , comes at the price of considerably larger N . Equation (13.2) indicates that halving f requires N four times as large.

Now let's be more realistic and assume that we don't know σ , and so our test statistic is t . Box 13.1 explains more about the formulas and tells us that, using t ,

$$N = [t_{.95}(N - 1)/f]^2 \quad (13.3)$$

where $t_{.95}(N - 1)$ is the critical value of t for a 95% CI. Note carefully that this N will give precision of target MOE *on average*, or in the long run, rather than in our single experiment. Figure 13.2 shows near the bottom such values with the label **N using t** . These are the minimum N values that give CIs having on average our target MOE. In every case, N needs to be a little higher when we use t rather than z .

Choose $f = 0.5$ and use t , and Equation (13.3) tells us that $N = 18$ is required for our average MOE to be no more than 10, our target MOE. That's also the value reported as **N using t** in Figure 13.2 under the leftmost CI. Take a sample of 18 students, test them on the HEAT, and calculate the 95% CI: MOE may be more or less than 10, but repeat the experiment numerous

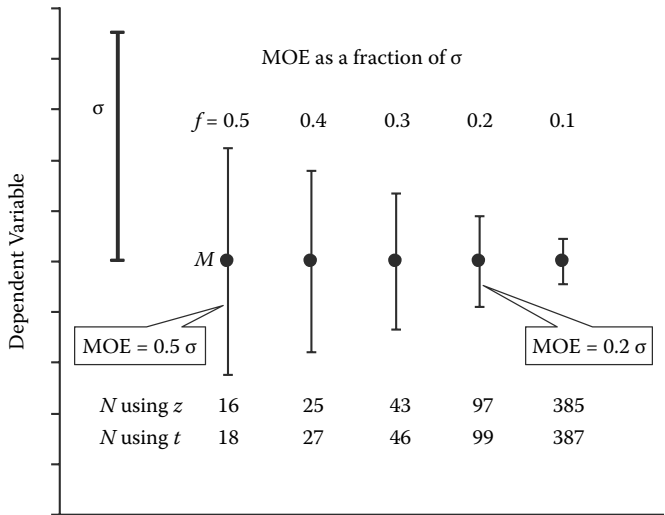


FIGURE 13.2
The line segment at left indicates the population standard deviation, σ . Then 95% CIs on M are shown with MOE of various fractions f of σ , as shown at the top. Using z , N using z is the size of a single sample that gives the CI in the figure. Using t , N using t is the size of a single sample that gives, *on average* over many experiments, the CI in the figure. (In each case a slightly shorter CI may be given, because N has been rounded up to an integer.) As Equations (13.2) and (13.3) state, the values of N depend only on f , not on σ .

times and, in the long run, the average MOE will be 10, or slightly less because N has been rounded up to an integer.

Note that the amount of bouncing around of CI arm length would be very different for small and large samples. For the smallest CI shown in Figure 13.2, which requires N of 387, there would be hardly any variation of MOE over replication, whereas with N of 18 there would be considerable variation. For small samples, the N using t in Figure 13.2 will give average MOE no more than our target MOE, but any single experiment could give MOE considerably bigger or smaller than our target.

There's one troubling thing about Equation (13.3). The value of $t_{.95}(N - 1)$ we need on the right-hand side of the equation depends on N , which is what we are trying to calculate. Change N , and $df = (N - 1)$ changes and the critical value $t_{.95}(N - 1)$ also changes.

If σ is not known, t is the test statistic. Use Equation (13.3) and an iterative approach to find minimum N that gives *on average* a particular target MOE.

We need to take an iterative or trial-and-error approach to solving Equation (13.3). We can leave the details to ESCI but, in outline, first use z instead of t to get an initial estimate of N , then check whether, using t , this N gives a sufficiently short MOE. If not, try $(N + 1)$ and again check MOE. Continue until N is sufficiently large to give MOE just less than our target MOE.

BOX 13.1 CALCULATING N GIVEN A TARGET MOE

We specify target MOE as $f \times \sigma$, as in Equation (13.1).

Single-Group Design

Back in Chapter 3 we calculated the 95% CI for a mean, when σ is known and z is the test statistic, as

$$[M - 1.96 \times \sigma / \sqrt{N}, M + 1.96 \times \sigma / \sqrt{N}] \quad (3.2)$$

Therefore, precision is

$$\text{MOE} = 1.96 \times \sigma / \sqrt{N} \quad (13.4)$$

More generally, for a $C\%$ CI using z , precision is

$$\text{MOE} = z_{C/100} \times \sigma / \sqrt{N} \quad (13.5)$$

where $z_{C/100}$ is the critical value of z for the $C\%$ confidence level. The **Normal z t** page of **ESCI chapters 1–4** can provide critical values for z and t (see Appendix B). Substitute $\text{MOE} = f \times \sigma$ in Equation (13.5) and find that

$$N = (z_{C/100}/f)^2 \quad (13.6)$$

which for $C = 95$ becomes

$$N = (1.96/f)^2 \quad (13.2)$$

Equation (13.6) is the basic formula for precision for planning with the single-group design, with σ assumed known. It's what the **Precision one** page of **ESCI chapters 10–13** uses to calculate N when using z .

For σ not assumed known and t as the test statistic, back in Chapter 3 we calculated the 95% CI by using Equation (3.3), which gives

$$\text{MOE} = t_{C/100}(N-1) \times s / \sqrt{N}$$

where $t_{C/100}(N-1)$ is the critical value of t for the $C\%$ confidence level. Substitute $\text{MOE} = f \times \sigma$ and find that

$$N = [t_{C/100}(N-1)/f]^2 \times (s/\sigma)^2$$

On average, $(s/\sigma)^2 = 1$, so we can calculate

$$N = [t_{C/100}(N-1)/f]^2 \quad (13.7)$$

which for $C = 95$ becomes

$$N = [t_{.95}(N - 1)/f]^2 \quad (13.3)$$

providing we note carefully that this N gives target MOE *on average*. (Or slightly less because N is rounded up to an integer.) Any single experiment may give precision larger or smaller than target MOE. For very large N , there is little variation in MOE with replication, but small N gives large variation in MOE from experiment to experiment. Think of the s pile in Chapters 10 and 11. Even N up to 50 or more gives noticeable variation in MOE with replication.

Because N appears on the right-hand side in Equations (13.3) and (13.7)—it affects the critical value of t —as well as the left, we need an iterative strategy of successive approximations to solve either of those equations for N . **Precision one** uses Equation (13.7) and an iterative approach to calculate N when using t .

Two-Independent-Groups Design

For the two-independent-groups design, our focus is on the difference between the group means and the CI on that difference. We assume that the two groups are each of size N . Using z , the equation for N that corresponds to Equation (13.6) is

$$N = 2(z_{C/100}/f)^2 \quad (13.8)$$

which is the smallest N that gives arm length of the CI on the difference between the sample means, calculated using z , no greater than target MOE. The **Precision two** page of **ESCI chapters 10–13** uses Equation (13.8) to calculate N when using z .

Using t , the equation for N that corresponds to Equation (13.7) is

$$N = 2[t_{C/100}(2N - 2)/f]^2 \quad (13.9)$$

which is the smallest N that gives, *on average*, arm length of the CI on the difference between the sample means, calculated using t , no greater than target MOE. **Precision two** uses Equation (13.9) and an iterative approach to calculate N when using t .

Paired Design

For the paired design, N is the number of data pairs and our focus is on the mean of the differences, and the CI on that mean. Using z , the equation for N that corresponds to Equations (13.6) and (13.8) is

$$N = 2(1 - \rho)(z_{C/100}/f)^2 \quad (13.10)$$

where ρ is the correlation in the population between the two measures. Equation (13.10) gives the smallest N so arm length of the CI on the mean difference, calculated using z , is no greater than target MOE. The **Precision paired** page of **ESCI chapters 10–13** uses Equation (13.10) to calculate N when using z .

Using t , the equation for N that corresponds to Equations (13.7) and (13.9) is

$$N = 2(1 - \rho)[t_{C/100}(N - 1)/f]^2 \quad (13.11)$$

which is the smallest N that gives, on average, arm length of the CI on the mean difference no greater than target MOE. **Precision paired** uses Equation (13.11) and an iterative approach to calculate N when using t .

ESCI for Precision for the Single-Group Design

Fire up the **Precision one** page of **ESCI chapters 10–13**. Figure 13.3 shows part of the control area. Figure 13.4 is the main display, whose curve shows how N varies as a function of target precision, using z . If $C = 95$ is set near red 1, the values of N shown on the curve correspond to the **N using z** values shown at the bottom in Figure 13.2. Figure 13.4 shows a vertical cursor positioned at a target MOE of $f = 0.2$ units of σ . This cursor is controlled by the large slider that appears on the screen below the figure. The value of N marked by the cursor is 97, which is the value shown for $f = 0.2$ in Figure 13.2, using z .

The vertical axis in the main figure needs to span a very wide range of N values. Adjust the slider near red 1 that's labeled **Truncate display at**, as shown in Figure 13.3, and the curve disappears for lower values of f . The vertical axis rescales automatically to display a smaller range of N values. Figure 13.5 shows the figure truncated at $f = 0.4$, meaning that 0.4 is the minimum value of f for which the curve is displayed. The curve for N using t is also displayed. Check that the values displayed correspond with those in Figure 13.2 for **N using t** .

Choosing Target Precision

How should we choose our target MOE? For our HEAT experiment we chose 10 score units as our target MOE. That was for the single-group case, but the discussion in this section applies also for other situations,

1 Sample size figure

Display N curves
☒ Using z
☐ Using t

Level of confidence
 C 95

☒ Display values
☐ Display values

Truncate display at: f 0.1

2 Assurance in figure

Level of assurance
99

☐ Display N curve for assurance
☐ Display values

3 Cursor for target MOE

☒ Display MOE cursor
☒ Display values

Target MOE cursor: f 0.2

5 Find N , for target MOE in original units

Target MOE 10

Population σ 20

☐ Display crosses

Target MOE in SD units: f 0.5

N 16 Using z
 N 18 Using t
 N 29 with assurance

FIGURE 13.3

Part of the control area of the **Precision one** page of **ESCI chapters 10–13**. The checkboxes near red 1 turn on the curves that show how N varies as a function of target MOE. Click at red 3 to turn on a vertical cursor to mark target MOE, which is set by a large slider below the main figure that appears on the screen to the right of the control area.

including the other two designs we’re considering in this chapter. We chose 10 by using our judgment in the research situation, based on understanding of the HEAT scale and differences in HEAT scores likely to be of research interest. We expressed that target MOE of 10 original units as $f = 0.5$, meaning 0.5σ . We could do that because we were assuming that we knew $\sigma = 20$. In many situations, however, we don’t know σ and use t as the test statistic. Even so, we can still express target MOE as f , in terms of the unknown σ . If we then choose N given by the formulas in Box 13.1 or by one of the ESCI precision pages, we know that, on average, MOE calculated from our data will be no greater than $f \times \sigma$. That will be true, even though we don’t know σ and so can’t express target MOE = $f \times \sigma$ in original units. After we’ve run our experiment we shift attention from target MOE and simply report the CI calculated from our data, then use this to guide our interpretation. We could also, if we wish, use s from our experiment to estimate σ . When thinking about the *next* experiment, we would, of course, take into account the extra knowledge we now have—about σ

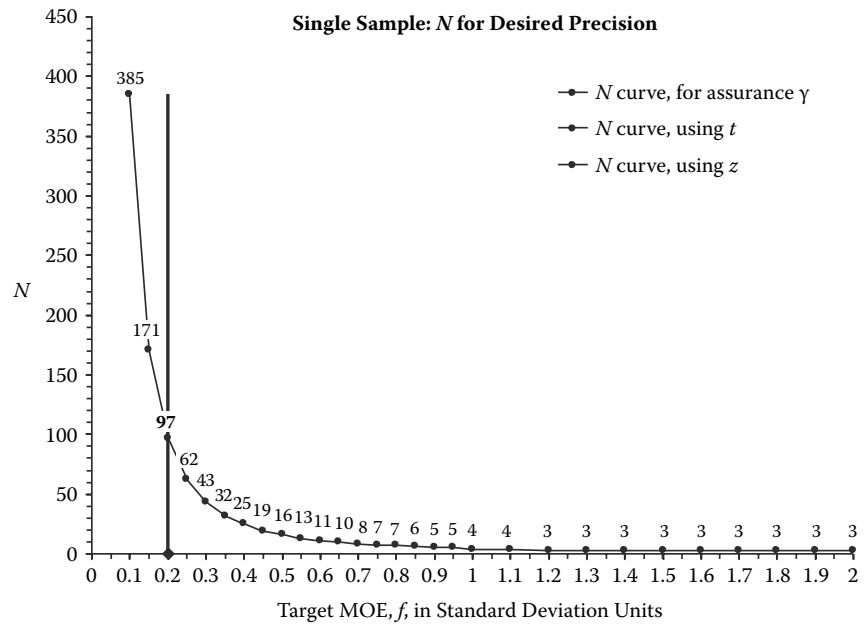


FIGURE 13.4

The main figure of **Precision one**. The curve shows N required to give target MOE for a 95% CI, using z as the test statistic. On the screen this curve is black. In this figure, and the following figures like it, the three legend entries at upper right are distinguished on the screen by color.

as well as the population ES—as we set target MOE to guide our choice of N for that next experiment. Choosing the f for our target MOE is still a matter for judgment, but it’s now better-informed judgment.

One way to think about f for planning is in general terms as a fraction of the unknown population SD, much as in Chapter 11 we discussed values of d or δ in numerous contexts as a number of SDs. In either case we might think in terms of Cohen’s benchmarks for small, medium, and large ESs, or use our experience in the research context to focus on a relevant fraction of population SD—even if we can’t express that σ exactly in original units.

Another approach is to find an estimate for σ , perhaps based on past research—preferably a meta-analysis—and use that as the SD needed to convert our chosen target MOE from original units to f . In Chapter 11 we discussed a range of approaches to finding such a standardizer. ESCI can do the arithmetic: In the panel at red 5, which is shown in Figure 13.3, you can enter your target MOE and a value for known or estimated population SD, both in any original units. The figure illustrates our case where we chose 10 as the target MOE and 20 as the population SD, both in HEAT scale units. ESCI calculates $f = 0.5$ and reports that N is 16 using z , and 18

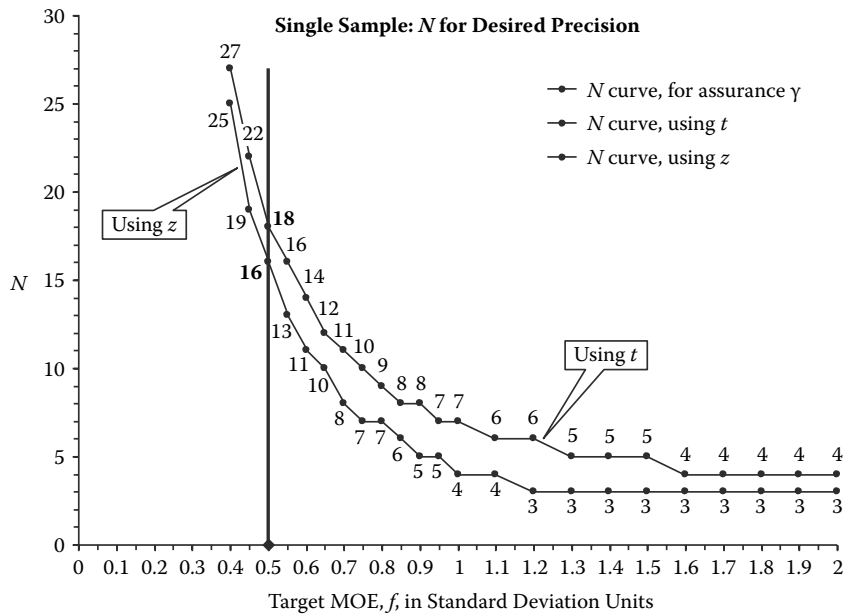


FIGURE 13.5
The main figure of **Precision one**, as in Figure 13.4, except that the curve for N using t is also displayed, and the curves are truncated at $f = 0.4$. The vertical N axis rescales accordingly. On the screen the lower curve, using z , is black, and the upper curve, using t , is red. The cursor marks $f = 0.5$ and shows $N = 16$ for z , and $N = 18$ for t .

using t , as we found before. Note that this panel is quite separate from the main figure. It's simply a small self-contained calculator that allows you to find N for values of target MOE that may not be displayed in the main figure. As usual, see the popout comments for further explanation.

You might be thinking that this is all rather subjective and there could be scope for fudging to get a value of N that we find acceptable. That may be true to some extent, depending on how well the research area and the measures we are using are understood. But, after we've run our experiment, the CI calculated from our data gives us precision information directly, and may give useful guidance for planning the next experiment.

The great advantage of precision, compared with power, is that the target MOE we use for planning corresponds closely to MOE of the CI we report after completing the experiment. Both may be in original or standardized units. Both require, for interpretation, our understanding of the measurement scale in the research context. In both cases the examples and discussion of standardizers in Chapter 11 may be relevant. We use our knowledge and judgment to interpret the reported CI, and that's a key part of interpreting our results. As our understanding of the measurement scale and research context develops, our interpretation of the CIs

that express our results improves. The same increased understanding can inform our precision analysis for planning.

Choice of target MOE is an important topic for discussion as researchers experiment with using precision for planning. I hope examples are published, and conventions and guidelines developed, but the important point is that improvement in how researchers use precision for planning should naturally accompany—and contribute to—improvement in researchers' interpretations of ESs and CIs based on data.

Finding N for the Two-Independent-Groups Design

For the two-independent-groups design I consider only cases in which $N_1 = N_2$, and so we have two groups each of size N . The focus is on the difference between the group means and the CI on that difference. Box 13.1 provides Equations (13.8) and (13.9) for this design, which correspond to Equations (13.6) and (13.7) for the single-group case. Once again, when using t an iterative approach is required, and N is the value that *on average* gives target MOE—or a little less because N is rounded up to an integer.

The ESCI page is **Precision two**, which is very similar in layout to **Precision one**. Figure 13.6 shows its main figure, with truncation at 0.45 and the vertical cursor positioned at $f = 0.5$. Compare with Figure 13.5 and note, for example, the N values marked by the vertical cursor when using z . A single-group experiment needs one group of size $N = 16$ to give a 95% CI on the sample mean with an arm length of 0.5σ . By contrast, a two-independent-groups experiment needs groups each with $N = 31$, a total of 62 participants, to give a 95% CI on the difference of the two means with that same arm length.

Finding N for the Paired Design

For the paired design, our focus is on the mean of the differences and the CI on that mean difference. We need to specify a value for ρ , the population correlation between the two measures, and we might look in the literature or rely on our judgment to choose a value for ρ that seems reasonable for the research situation. In Chapter 12 we found that the value of ρ is very influential on power. It's also very influential on precision: Higher ρ means that a much smaller N is needed to achieve our desired precision. Yes, the paired design is sensitive and efficient, for situations in which it is appropriate, and especially so if the two measures are highly correlated.

Box 13.1 provides Equations (13.10) and (13.11) for the paired design. Once again, when using t an iterative approach is required, and N is the value that *on average* gives target MOE—or a little less because N is rounded up to an integer.

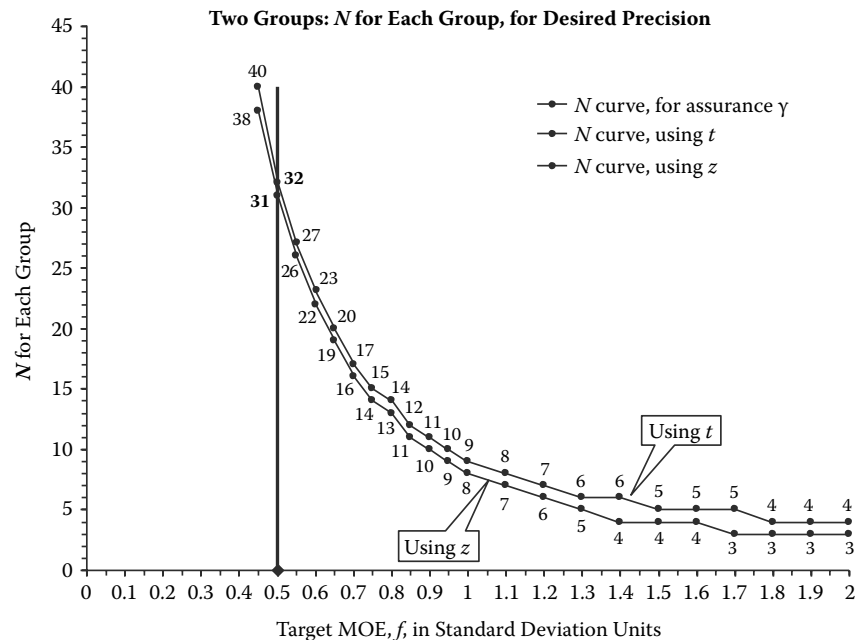


FIGURE 13.6
The main figure of **Precision two**. The lower curve shows N for z , and the upper curve shows N for t , for two independent groups each of size N . The lower axis shows target MOE for a 95% CI on the difference between the two group means.

The ESCI page is **Precision paired**, and Figure 13.7 shows a small part of the control area of that page, including the slider that allows setting of a value for population correlation ρ between the measures. Change ρ and watch the marked changes in the curves of N against f , the target MOE. Figure 13.8 shows the main figure from **Precision paired**, when $C = 95$ and $\rho = .7$ have been set near red 1. With these settings, N of 10 for

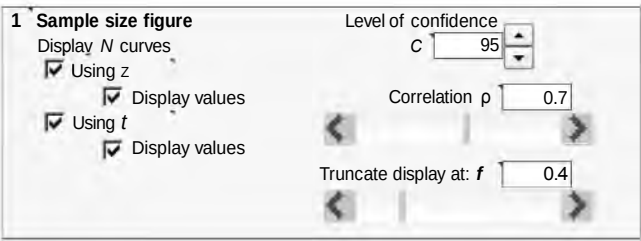


FIGURE 13.7
The top control panel of **Precision paired**. The upper slider allows the setting of a value for ρ , the population correlation between the two measures.

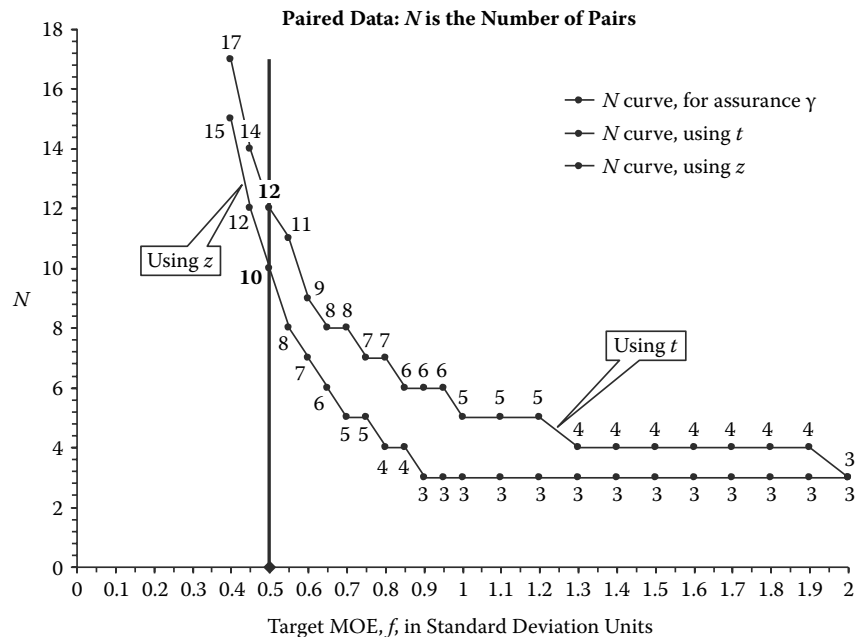


FIGURE 13.8
The main figure from **Precision paired**. The lower curve gives N using z , and the upper gives N using t . The curves relate to 95% CIs and $\rho = .7$.

z and 12 for t are sufficient to achieve the target $f = 0.5$. (For t , have you automatically added the mental comment “on average, in the long run”?) Increase ρ and even smaller N will suffice. Alternatively, aim for greater precision—meaning smaller target MOE—and find that even then N need not be large. Adjust the lower slider at red 1 to shift the truncation of the curves and reveal N for smaller values of f . Note that $N = 3$ is the smallest N that ESCI permits in any precision calculations.

Precision with Assurance

Use t as the test statistic, as we usually must, and so far we can only calculate N that guarantees an arm length that *on average* isn’t too large. It’s a concern that almost half of the experiments conducted with that N will give MOE greater than our target. The next step is to calculate N so there’s a 99% chance that our sample MOE will not exceed the target MOE. The 99% is one possible value of what’s called *assurance*, which I’ll give

the symbol γ (Greek lowercase gamma). Assurance γ is the probability, expressed as a percentage, that our sample MOE is no greater than target MOE. We can choose $\gamma = 99$, or $\gamma = 80$ (the lowest value ESCI supports), or some other value less than 100. (Some authors use the term “certainty,” but “assurance” is becoming the most widely used term and is my preference.) I don’t know of any convention to guide our choice of γ , but I’m inclined to choose $\gamma = 99$ so that my MOE is rarely larger than target MOE. I try to avoid $\gamma = 95$ because that may encourage confusion between γ and C , the level of confidence.

Assurance γ is the percentage of experiments whose MOE is less than target MOE.

You’ve probably noticed panel 2, which appears on each of the precision pages and in Figure 13.3. Panel 2 provides a spinner to set assurance γ . You can also click to see an additional curve in the figure, which appears above the z and t curves, and gives N for the selected value of γ . Figure 13.9 shows an example from **Precision one**. It’s the same as Figure 13.5 except a third curve now gives N with assurance $\gamma = 99$. Using t as our test statistic, we noted before (and saw in Figure 13.2) that $N = 18$ suffices to give, on average, MOE of 0.5σ . Read the value of N where the vertical cursor

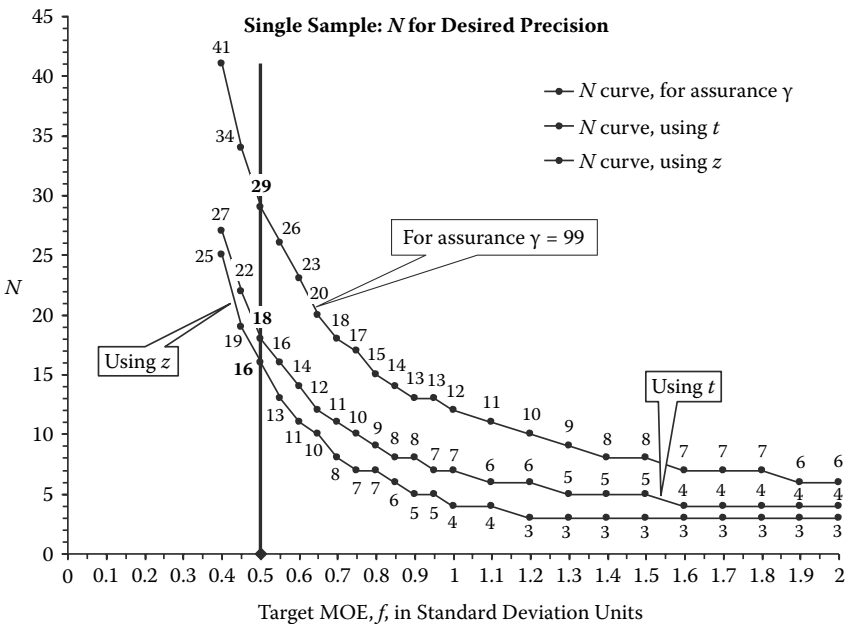


FIGURE 13.9 Same as Figure 13.5, for a single group, but with the curve for N with assurance $\gamma = 99$ also displayed. On the screen, the assurance curve is pink.

BOX 13.2 CALCULATING N WITH ASSURANCE

When we use t , assurance γ is the percentage of experiments whose MOE is no greater than target MOE. In earlier chapters I mentioned that s^2 has a sampling distribution with the shape of the chi-square distribution. Think also of the s pile, shaped like a chi distribution, which has a single hump and positive skew. When s (or s^2) happens by chance to be large—to be in the upper tail of its sampling distribution—the CI we calculate for that experiment is especially large. We can use knowledge of the sampling distribution of s^2 to calculate how much larger N needs to be so that only a small upper tail of that distribution, with area $(1 - \gamma/100)$, gives a CI that is wider than we want. That's the basis for assurance calculations.

I'm not going to explain those calculations in detail, but the basic formula for the single-group case is

$$N = [t_{C/100}(N-1)/f]^2 \times [\chi^2_{\gamma/100}/(N-1)] \quad (13.12)$$

where $\chi^2_{\gamma/100}$ is the critical value of χ^2 for our selected γ , with $df = (N-1)$. Note that N appears on the right-hand side of the equation, so, as before with t , we need a step-by-step, successive approximations procedure to find the smallest N that will satisfy our requirement for MOE to be shorter than target MOE on at least $\gamma\%$ of occasions. As usual, ESCI takes care of those details.

intersects the top curve in Figure 13.9, and see that $N = 29$ is needed if we are to be 99% assured that our MOE will be no more than the target $f = 0.5$. We need a considerably larger sample to gain 99% assurance that our MOE won't be longer than we want. Box 13.2 says a little about the calculation of N with assurance.

I suggest that precision calculated with assurance is highly useful, and probably should be our routine practice for finding N during the planning of experiments. We can't be absolutely guaranteed we won't get a CI longer than we want, but select $\gamma = 99$ and we can be pretty sure. Of course, if we use the larger N that's needed to achieve that degree of assurance, our expected MOE will be shorter. Figure 13.9 shows that $N = 27$ gives an expected target MOE of $f = 0.4$, using t . Therefore, if we use $N = 29$, on average our MOE will be a little shorter than $f = 0.4$. Also, on 99% of occasions our MOE will be shorter than target $f = 0.5$, so on only 1% of occasions will it be longer than that target. In summary, on average our MOE will be about 0.4σ and only rarely will it be greater than 0.5σ : That's good information about the precision that our planned experiment will achieve.

Precision Beats Power

Compare precision with power. From Chapter 5 we know that the p value has an extremely wide distribution and so, in advance of our experiment we have very little idea indeed of what p value we'll get, even if target δ used in the power calculation equals δ in the population. All a power analysis can tell us is that, assuming that target δ equals population δ , we have an 80% chance of obtaining $p < .05$ —if power is .8 and we're using $\alpha = .05$. In addition, even for very large N , the distribution of p is just as wide, so the uncertainty about our obtained p is just as great, no matter how large N is. In stark contrast, using precision for planning has at least three advantages:

1. It gives us information about arm length, which should be readily interpretable in the research context, and corresponds directly with the CIs we use for interpretation of our results.
2. With an assurance analysis, the information about arm length is fairly detailed, as I described previously: We know both an expected MOE and a likely upper bound.
3. The extent of bouncing around of arm length decreases for larger N , so the uncertainty in a precision analysis, unlike a power analysis, steadily decreases with increasing N .

Yes, precision beats power.

In my assurance example above, N needed to be increased from 18 to 29 when we required 99% assurance that our MOE was no longer than our target $f = 0.5$. That's a 61% increase in N . It's large because, for the particular situation and sample size, CI length bounces around considerably with replication. That observation accords with a general theme of this book that, unfortunately, in many situations the amount of variation over replication is large, even surprisingly large. Think of the dance of the p values. However, with assurance there's good news. For many typical cases that arise in practice, the increase in N required to achieve high assurance is small. For example, use **Precision two** and set a target MOE of $f = 0.3$. For 95% CIs, N needs to be increased from 87 to 108 to achieve assurance of $\gamma = 99$, a more reasonable increase of 24% in sample size. Explore different designs and various targets for MOE and find that the increase in N needed to achieve high assurance can often in practice be modest. Kelley and Rausch (2006, pp. 375–379) discussed this issue in the context of an example in which achieving $\gamma = 99$ required an increase of only 7% in N . Using assurance as well as precision to plan experiments may be one of the rare situations in which sampling variability is often not surprisingly

large. Consider also what peace of mind $\gamma = 99$ can give: We're almost guaranteed that our experiment will give a CI no longer than the target used in the assurance analysis.

Precision for Planning Using Cohen's d

This chapter has so far referred to means, but now I'll consider Cohen's d . What N do we need for expected MOE_{av} of the CI on our d to be no more than some target? (Recall from Figure 13.1 that MOE_{av} is the average of the two arms of an asymmetric CI. By "expected MOE_{av} " I'm referring to the long-run average MOE_{av} over many experiments.) That's possible to calculate, but tricky. Recall that to find the CI on an observed d , we need an iterative (i.e., successive approximations) procedure that uses noncentral t . Chapter 11 tells the story, and the **CI for d** page of **ESCI chapters 10–13** can do the work. If we have a two-independent-groups experiment with $N = 20$ for both groups, and observe $d = 0.45$, **CI for d** calculates the 95% CI for δ to be $[-0.18, 1.08]$ and $\text{MOE}_{\text{av}} = 0.63$. Yes, quite a wide CI, so perhaps we should be using larger N ?

You may have noticed that the three precision pages in ESCI have a panel at red 6 for calculations for Cohen's d . This d panel is a self-contained calculator. It uses values of C and γ set in the control panels above it, but has no links with the main figure. Figure 13.10 shows this panel for **Precision one**. The panel for **Precision two** is very similar. Set C near red 1 and γ near red 2, and then use the spinner in the upper right of this panel to set your desired target MOE_{av} . This, like f , is a number of population

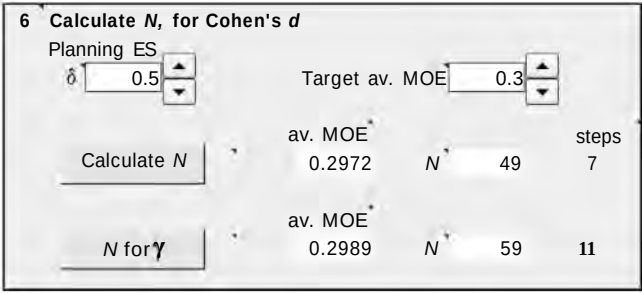


FIGURE 13.10
The control panel for Cohen's d from **Precision one**. In the panels above this one, as shown in Figure 13.3, $C = 95$ and $\gamma = 99$ have been set. This figure shows results for N after both buttons have been clicked.

BOX 13.3 PRECISION AND ASSURANCE CALCULATIONS FOR COHEN'S d

My calculations for d are partly based on the work of Kelley and Rausch (2006). The idea is that we start with a guess for N and use the iterative procedure also used by the **CI for d** page of **ESCI chapters 10–13** to find expected MOE_{av} . If this is larger than the target MOE_{av} , increase N and try again. If it's smaller, decrease N . Stop when N is the smallest that gives expected MOE_{av} that's not greater than the target. That brief version skips over some details, notably how we calculate expected MOE_{av} . But the general picture is that we need two nested iterative or successive approximation procedures, the inner one to find the expected MOE_{av} for a given N , and the outer one to try different values of N until we home in on the value we want. Finding N for a particular assurance complicates the picture further, but the basic idea of nested iterations still applies.

SDs, or, most often, a fraction of the population SD, which is σ . Note that d and δ are also expressed as a number of population SDs, so we can think of the target MOE_{av} as a number expressed on the same scale as d and δ . Next, click the **Calculate N** button to trigger calculation of the smallest N that gives on average in the long run MOE_{av} no larger than your target. Box 13.3 describes a little about the calculations, which require a double iteration process that may take some time, perhaps a minute or more. As ESCI labors away, you can watch as successive MOE_{av} and N values are shown to the right of the button. At the far right a counter reports the number of main iterative steps taken so far. If ESCI cannot complete the calculation, **No** is shown instead of a value of N .

Similarly, you can click the **N for γ** button and watch as ESCI finds N for the assurance level γ you have set. When either process successfully gives a value of N , the value shown as **av. MOE** just beside the button should be equal to, or a little less than, the MOE_{av} you set as target. As usual, see the popout comments for further details, including limits on the values that ESCI can handle.

In the top left of the panel is a spinner that sets a value labeled **Planning ES** and δ . It's an added complication that an assumed value of population δ is required for use in the calculations. The value you set for planning δ makes a difference, but not an enormous difference. Figure 13.10 shows that N values of 49 (using **Calculate N**) and 59 (using **N for γ**) are needed when $\delta = 0.5$ is the planning value, and other settings (C , γ , and target MOE_{av}) are as described above. Change δ to 0.2 and find the N values become 44 and 50. Change δ to 0.8 and the N values become 57 and 72.

Yes, the planning value you set for δ matters, but the N values I've just described relate to δ values ranging from small (0.2) to large (0.8), which in practice is a very wide range. The differences here in N are nothing like the dramatic differences in power that even a much smaller change in target δ gives. In practice, you usually have a rough idea of the likely δ in your research situation, and can use that as your planning value for δ . To be conservative (and get slightly larger N values), err on the high side when setting your planning δ . Of course, you can try out more than one planning δ .

The **Precision paired** page also includes a panel for Cohen's d . In Chapter 11 I described the approximation I used to calculate CIs on d for the paired design, and the limitations of its applicability. Those limitations apply here also. Our C must be 95 (set that value near red 1) and planning δ can be no larger than 2. The N we calculate for our nominated target MOE_{av} will be an approximation, but it's a reasonably accurate approximation, sufficiently accurate for practical purposes. You'll notice that there's no second button inviting you to click to carry out an assurance calculation. I haven't been able to find any published analysis of such a calculation for the paired design. So far as I'm aware, it remains an interesting problem awaiting a good solution.

The Future of Precision for Planning

I hope precision for planning will become more widely used. I regard precision and assurance calculations for planning as having great value, even though they have to date been seldom used by researchers. They offer quantitative guidance for research planning within the world of estimation, while avoiding the baggage and disadvantages of statistical power. There's a close and natural link from a planning analysis that considers target precision to the reporting of results using CIs—which tell us the precision of our findings. I would like to include published examples of researchers reporting their precision calculations to illustrate the practical usefulness of precision and assurance for planning. Unfortunately, I haven't been able to find suitable examples. It remains a somewhat pioneering exercise to use precision and assurance for planning, but please be encouraged to try. I hope the exercises at the end of this chapter help.

The *Publication Manual* (APA, 2010) acknowledges precision analysis as part of planning. It recognizes that an alternative to power calculations are "calculations based on a chosen target precision (confidence interval width) to determine sample sizes" (p. 31). It also refers to "how sample size

was determined ... [based on] precision of parameter estimates" (p. 248). That's important legitimation of the approach.

I used only means and Cohen's d in my discussion of precision for planning. However, we could use correlations, proportions, odds ratios—or any other ES measure appropriate for our research—although considerable further statistical development is needed before we will have proven methods for all the ES measures researchers wish to use, and easily accessible software to do the work. Precision hasn't yet received sufficient attention, and assurance has received even less. I'm hoping that wider use of the new statistics will prompt rapid developments. It will be great to have convenient software to easily carry out any precision or assurance analysis we wish, to inform our research planning. Meanwhile, back on planet Earth ...

Actually, there is hope. Researchers are publishing new techniques for precision calculations for various situations, and also investigations of the accuracy and applicability of those techniques. Much of this research is labeled "accuracy in parameter estimation" (AIPE). I mentioned before the work of Kelley and Rausch (2006), which guided my calculations using d . More recently, Maxwell, Kelley, and Rausch (2008) reviewed AIPE and compared and contrasted the power and AIPE approaches to planning.

When using precision and assurance for planning, it may be useful to take an exploratory approach and get a feel for how N and f trade off, for various experimental designs in your research situation. In each of the ESCI pages for precision, the main figures show how N varies with target f , for means. The figures make it easy to investigate what happens for various values of f and γ , and to compare different designs. The calculations for Cohen's d take longer, and it's not feasible for ESCI to display curves showing how N relates to target MOE_{av} , but it can still be valuable to explore variations. Investigate how N changes as you set different values for MOE_{av} , and different values for planning δ . This may help build intuitions about experiments you could run in your own research context.

Ways to Increase Precision

In Chapter 12, I distinguished statistical power from the informativeness of an experiment. I also need to distinguish informativeness from precision. Informativeness is a very general concept, whereas precision refers specifically to the size of estimation error. The most general way to increase informativeness is to apply ingenuity and creativity—and brainstorming, consultation, deep thought, and the exploration of possibilities—to improve the match between the planned experiment and the research problems it addresses. What experiment will give us the greatest

insight into the way the universe works? That's what really matters. More prosaically, increasing the precision of the estimates an experiment gives is likely, in many situations, to improve informativeness. I mentioned in Chapter 2 the call by Paul Meehl (1978) for a shift from NHST and a focus on building a more quantitative discipline of psychology. Increasing precision is likely, in many situations, to contribute to that worthy goal.

When I listen to graduate students presenting their research plans, I'm often impressed with their logic and their care to choose appropriate manipulations and experimental design. However, I'm often dismayed at their optimism that the effects they are so passionate to investigate will actually emerge. In other words, their ingenuity has often identified experiments that potentially are highly informative, but which lack precision, and therefore risk being uninformative. The problem is not confined to students—witness the continuing evidence that much published research typically has low power (Maxwell, 2004).

A vital part of any research planning is repeated brainstorming about how to increase informativeness. Usually, some of that effort should be devoted to improving precision, and it's this aspect I'll focus on here. Gather colleagues or fellow students, and spend time trying to find ways to increase precision. Here are a few suggestions:

- Consider a more sensitive design. Matching? Repeated measures?
- Can you increase p in a paired design, perhaps by better matching?
- Increase sample sizes.
- Reduce error of measurement. Consider measuring more than once and averaging.
- Find better measures, likely to have higher reliability and validity.
- Can meta-analysis usefully combine results from several of your studies, or from several outcomes of a large study?

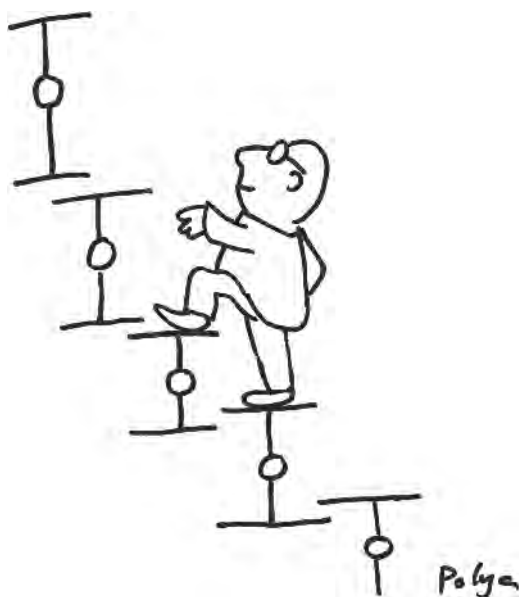
You should be able to extend that list considerably. I'll end by repeating my main point. Calculation of precision and assurance can be a very useful part of research planning, and I hope they become widely used and appreciated. However, they should inform, not replace, careful thought, creativity, ingenuity, and judgment about what experiment is likely to be most informative in your research situation.

It's time again for take-home messages. I started this chapter by defining precision as CI arm length. Precision is better than power. Turn around the formula for a CI and we can calculate N likely to give a particular target precision. Three experimental designs, and three precision pages in ESCI. Assurance as well as precision. Cohen's d . How to increase precision, and what's really important.

Exercises

- 13.1 Revisit some discussions of CIs, or figures of CIs, in earlier chapters, and recast the discussion, or describe the pictures, in terms of precision.
- 13.2 Scan any other statistics textbook you are using for any mention of precision. Does it arise as a concept, even by some other name, in connection with calculating or reporting CIs? In connection with interpreting CIs? As part of research planning? In any other way?
- 13.3 If you, or some friends or colleagues of yours, have knowledge about physics, chemistry, engineering, or some other technical discipline, investigate what's meant in that discipline when you see a measurement reported as, for example, 18.5 ± 0.1 . What's meant by the " ± 0.1 " value, and how is it defined and measured? Is there any link to precision as I use the term?
- 13.4 Could we have a "dance of the precision values"? If so, what use might it have, and how would it relate to the various other dances we've encountered?
- 13.5 Refer back to Exercise 12.1. In that situation, suppose you would like to achieve precision of $\text{MOE} = 0.2$ scale units, using z . What N would you need? Using t , what N would give you expected precision of that target value? What N would you need to have 90% assurance of obtaining that target precision?
- 13.6 Refer back to Exercise 12.2. In that situation, what N would you need to achieve precision of $\text{MOE} = 0.1$, using z ? Using t , if that's to be the expected precision? With 99% assurance?
- 13.7 Refer back to Exercise 12.3. Carry out a precision and assurance analysis, and discuss.
- 13.8 Refer back to Exercise 12.5. Carry out a precision and assurance analysis, and discuss.
- 13.9 Refer back to Exercise 12.7. Do the comparable things for precision.
- 13.10 Refer back to Exercise 12.8. Choose a few benchmark cases of N and precision to remember.
- 13.11 Refer back to Exercise 12.9. Do the comparable things for precision.
- 13.12 If you are using any other statistics textbook, try to find some worked examples or exercises for which the ESCI precision pages are relevant. Compare the answers given by ESCI and that textbook.

- 13.13 In your own discipline, try to find in journal articles some use, or even mention, of precision. Compare with the discussion in this chapter.
- 13.14 Extend the bullet point list of ways to increase precision that I gave near the end of this chapter.
- 13.15 Revisit your take-home messages. Improve them and extend the list if you can.



Take-Home Messages

- Precision is measured by the arm length of a CI and is referred to as MOE, the margin of error. For asymmetric CIs we use MOE_{avr} , which is the average of the two arm lengths of the CI. MOE varies with C , the level of confidence of the CI.
- Precision is the largest likely error of estimation, as indicated by MOE.
- Watch out for ambiguous language: High or large precision is indicated by short or small MOE, and low or poor precision by a long or wide CI.

- Adopting estimation, and reporting and interpreting CIs means that precision is being reported. Any CI can be interpreted in terms of precision—its arm length—as Interpretation 3 of CIs states. This can often be a valuable approach.
- For planning, precision is better than power and avoids any need to invoke NHST or specify α and null and alternative hypotheses. There is a strong correspondence between the target MOE used in planning and MOE of the CI calculated from data. Similar knowledgeable judgment is needed to set target MOE and, later, to interpret the obtained CI.
- Precision can have a valuable role in research planning. Turn around the formula for a CI, and calculate what N is required to achieve a CI that's not too long.
- If z is the test statistic, we can calculate N to achieve a target MOE. When using t , MOE varies with replication and so we calculate N to achieve a target expected MOE, meaning the average MOE given by an indefinitely large number of replications.
- When using t , we can also calculate N so there's a $\gamma\%$ chance that MOE is no more than a specified target MOE. Then γ is the assurance, and we may choose $\gamma = 99$, for example.
- The pages **Precision one**, **Precision two**, and **Precision paired** of **ESCI chapters 10–13** show, for those three designs, the relation between N and target MOE, which is expressed as f standard deviation units. *Take-home picture*: Three curves of N plotted against target MOE, for z , t , and assurance, as in Figure 13.9.
- Using Cohen's d as the ES, those ESCI pages can carry out precision calculations and assurance calculations for the single-group and two-independent-groups designs. For the paired design, an approximation must be used to calculate CIs on d , and only limited precision calculations can be carried out.
- Precision has so far been little used by researchers for planning, but has great potential. Further developments are needed in techniques, software, and practical guidance for researchers.
- A fundamental aim of research planning is to increase the informativeness of experiments. Increasing their precision can often contribute. Achieving higher precision is often worth great attention and effort.